

Task 12: Log Monitoring & Analysis

Tools: Linux logs, Windows Event Viewer

Alternatives: Splunk Free

Hints / Mini Guide:

- 1.Understand log types.
- 2.Analyze authentication logs.
- 3.Identify failed logins.
- 4.Detect anomalies.
- 5.Correlate events.
- 6.Learn SIEM basics.
- 7.Write alerts.
- 8.Document findings.

Understand Log Types

Logs are records generated by operating systems, applications, and network devices to track activities and events. Different log types serve different security purposes. System logs record OS-level events such as boot processes and hardware issues. Authentication logs capture login, logout, and privilege escalation activities. Application logs store errors and operational messages from software. Security logs are critical for detecting unauthorized access, policy violations, and potential cyberattacks. Understanding log types helps analysts know where to look when investigating incidents.

Analyze Authentication Logs

Authentication logs contain information about user login attempts and access control decisions. These logs record successful and failed authentication events for services such as SSH, sudo, and local user logins in Linux systems. Analyzing authentication logs helps security analysts verify legitimate access, detect unauthorized login attempts, and monitor user behavior. Regular analysis allows early detection of compromised accounts and misuse of credentials.

Identify Failed Logins

Failed login attempts occur when a user provides incorrect credentials or attempts to access a system without authorization. These events are important indicators of brute-force attacks, password guessing, or unauthorized access attempts. By monitoring repeated failed logins from the same user account or IP address, analysts can identify attack patterns and take preventive actions such as blocking IPs or locking accounts.

Detect Anomalies

Anomalies are activities that deviate from normal user behavior or system patterns. Examples include logins at unusual times, sudden spikes in authentication failures, or access attempts from unknown locations. Detecting anomalies helps uncover potential security incidents that may not trigger traditional alerts. Behavioral analysis of logs allows analysts to identify subtle threats such as insider attacks or compromised accounts.

Correlate Events

Event correlation involves connecting multiple related log entries across time or systems to form a complete picture of an incident. For example, several failed login attempts followed by a successful

login may indicate a successful brute-force attack. Correlation improves incident detection accuracy by reducing false positives and providing context, making it easier to identify complex attacks.

Learn SIEM Basics

SIEM (Security Information and Event Management) systems collect logs from multiple sources into a centralized platform. SIEM tools analyze, normalize, and correlate log data to detect security threats in real time. They provide dashboards, search capabilities, alerts, and reporting features. Learning SIEM basics helps security professionals monitor environments efficiently and respond quickly to incidents.

Write Alerts

Alerts are automated notifications triggered when predefined security conditions are met. Examples include excessive failed login attempts within a short period or unauthorized access to privileged accounts. Writing effective alerts involves defining thresholds, time windows, and risk levels. Proper alerting ensures timely response while minimizing false alarms, improving overall security monitoring.